

MODULE 1

HyperText Markup Language

Assignment – Questions & Answers

20 Questions

Module 1 – HTML Assignment Answers

1. Are HTML tags and elements the same thing?

No, HTML tags and elements are not the same thing, though they are closely related.

A tag is just the markup notation written inside angle brackets. For example, `<p>` is an opening tag and `</p>` is a closing tag. A tag by itself is just a single marker.

An element is the complete unit — it includes the opening tag, the content inside, and the closing tag together. For example, `<p>Hello</p>` is an element. Every element is made of tags, but a tag alone is not an element.

2. What are tags and attributes in HTML?

Tags are keywords enclosed in angle brackets (`<` `>`) that define how content is structured on a web page. Most elements have an opening tag and a closing tag (with a forward slash).

Attributes provide additional information about an HTML element. They are written inside the opening tag as `name="value"` pairs and modify the element's behaviour or appearance.

```
<a href="https://example.com" target="_blank">Visit Site</a>

```

Key points:

- Attributes always go inside the opening tag, never the closing tag.
- Format: `attributeName="value"`
- Global attributes work on any element: `id`, `class`, `style`, `title`
- Some are element-specific: `href` only on `<a>`, `src` on `<img>`
- Boolean attributes need no value, e.g., `<input disabled>`

3. What are void elements in HTML?

Void elements are HTML elements that cannot have any child content and do not have a closing tag. They insert something directly into the page and are complete on their own.

In HTML5, void elements do not need a trailing slash. `<br>` is correct. In XHTML, the slash is required: `<br />`

Common void elements:

```
<br>      -- Line break
<hr>      -- Horizontal rule / divider
  -- Image
<input type="text">  -- Input field
<link rel="stylesheet" href="style.css">
<meta charset="UTF-8">
<source>  <track>  <col>  <area>  <base>
```

4. What are HTML Entities?

HTML Entities are special codes used to display reserved characters (characters that have meaning in HTML) or characters that are hard to type. They start with & and end with a semicolon (;).

Without entities, characters like < and > would be misread as HTML tags by the browser.

Common HTML Entities:

< = < (less-than, reserved in HTML)
> = > (greater-than, reserved)
& = & (ampersand, reserved)
 = non-breaking space
" = double quote
© = copyright symbol
® = registered trademark
€ = euro currency symbol

```
<!-- Without entity: browser treats this as a tag -->
If a <b> tag is used...

<!-- With entity: renders as visible text correctly -->
If a &lt;b&gt; tag is used...
```

5. What are different types of lists in HTML?

HTML has three types of lists:

1. Ordered List (): Items are displayed in numbered sequence. Used when order matters, like steps in a process.
2. Unordered List (): Items are displayed with bullet points. Used when order does not matter.
3. Description List (<dl>): Used for term-definition pairs. <dt> defines the term and <dd> gives the description.

```

<!-- Ordered List -->
<ol>
  <li>Open the browser</li>
  <li>Type the URL</li>
  <li>Press Enter</li>
</ol>

<!-- Unordered List -->
<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>

<!-- Description List -->
<dl>
  <dt>HTML</dt>
  <dd>HyperText Markup Language</dd>
  <dt>CSS</dt>
  <dd>Cascading Style Sheets</dd>
</dl>

```

6. What is the 'class' attribute in HTML?

The class attribute assigns one or more CSS class names to an HTML element. It allows multiple elements to share the same styling or JavaScript behaviour.

Key points:

- The same class can be applied to any number of elements on a page.
- A single element can have multiple classes, separated by spaces.
- In CSS, a class is selected using a dot: `.classname { }`
- In JavaScript: `document.getElementsByClassName('classname')`
- The class attribute has no semantic meaning on its own.

```

<p class="highlight">This text is highlighted.</p>
<span class="highlight">So is this span.</span>

<!-- Multiple classes on one element -->
<div class="card featured large">Content</div>

```

7. What is the difference between the 'id' attribute and the 'class' attribute of HTML elements?

Both id and class are used to identify HTML elements, but they differ in uniqueness and how they are used.

id Attribute:

- Must be unique on the entire page. Only one element can have a given id.
- Used to identify a single specific element.

- CSS selector uses a hash symbol: `#header { }`
- Also used for in-page anchor navigation: `href="#sectionName"`
- Has higher CSS specificity than class.

class Attribute:

- Can be shared by multiple elements on the same page.
- Used to group elements that share styles or behaviour.
- CSS selector uses a dot: `.button { }`
- An element can have multiple classes at once.
- Has lower CSS specificity than id.

```
<div id="main-header">Unique Header</div>

<button class="btn btn-primary">Save</button>
<button class="btn btn-secondary">Cancel</button>

<!-- An element can have both -->
<section id="hero" class="section featured">...</section>
```

8. What are the various formatting tags in HTML?

HTML provides the following tags for formatting text:

- `<b>` - Bold text (visual only, no semantic meaning)
- `<strong>` - Important bold text (semantic, screen readers stress it)
- `<i>` - Italic text (visual only)
- `<em>` - Emphasized italic text (semantic)
- `<u>` - Underlined text
- `<mark>` - Highlighted text (HTML5)
- `<s>` - Strikethrough (content no longer relevant)
- `<del>` - Deleted text (semantic)
- `<ins>` - Inserted text (semantic)
- `<sup>` - Superscript, e.g. x squared
- `<sub>` - Subscript, e.g. H₂O
- `<small>` - Smaller text (fine print)
- `<code>` - Inline code snippet
- `<pre>` - Preformatted text (preserves spaces and line breaks)
- `<blockquote>` - Block quotation from an external source

```
<p>This is <b>bold</b> and <i>italic</i> text.</p>
<p>Water is H<sub>2</sub>O and energy is E=mc<sup>2</sup></p>
<p><mark>This is highlighted.</mark></p>
<p><del>Old price: $50</del> New price: $30</p>
```

9. How is Cell Padding different from Cell Spacing?

Both are attributes of the HTML `<table>` tag but control different kinds of spacing.

Cell Padding:

The space between the content inside a cell and the cell's own border. It is like an inner margin within each cell. Set using the cellpadding attribute.

Cell Spacing:

The space between adjacent cells, and between cells and the outer table border. It controls the gap between cells. Set using the cellspacing attribute.

```
<table border="1" cellpadding="10" cellspacing="5">
  <tr>
    <td>Cell A</td>
    <td>Cell B</td>
  </tr>
</table>

/* Modern CSS equivalents */
td { padding: 10px; }          /* replaces cellpadding */
table { border-spacing: 5px; } /* replaces cellspacing */
table { border-collapse: collapse; } /* removes all spacing */
```

10. How can we club two or more rows or columns into a single row or column in an HTML table?

HTML provides two special attributes to merge cells across multiple rows or columns:

colspan (Column Span):

Merges a cell horizontally across multiple columns. The value indicates how many columns it should occupy.

rowspan (Row Span):

Merges a cell vertically across multiple rows. The value indicates how many rows it should span.

Important: When using rowspan="2", the next row must have one fewer <td> because that column space is already occupied by the spanned cell.

```
<table border="1">
  <tr>
    <td colspan="2">Merged across 2 columns</td>
  </tr>
  <tr>
    <td rowspan="2">Spans 2 rows</td>
    <td>Row 1 Data</td>
  </tr>
  <tr>
    <!-- Only 1 td here, first column is occupied by rowspan -->
    <td>Row 2 Data</td>
  </tr>
</table>
```

11. What is the difference between a block-level element and an inline element?

Block-level Elements:

- Always start on a new line.
- Take up the full width available.
- Can contain both block and inline elements.
- Have top and bottom margins by default.
- Examples: <div>, <p>, <h1> to <h6>, , , <table>, <section>, <article>

Inline Elements:

- Do not start on a new line; they flow within surrounding text.
- Only take up as much width as their content needs.
- Should only contain other inline elements or text.
- Top and bottom margins do not apply by default.
- Examples: , <a>, , , , <input>, <label>

Note: The CSS display property can override this default behaviour.

```
<p>Paragraph one.</p>
<p>Paragraph two starts on a new line.</p>

<p>This is <strong>bold</strong> and <em>italic</em>
on the same line without breaking flow.</p>
```

12. How to create a Hyperlink in HTML?

A hyperlink is created using the <a> (anchor) tag with the href attribute pointing to the destination.

Key attributes:

- href : The URL or path the link points to (required)
- target : Where to open the link (_blank = new tab, _self = same tab)
- rel : Relationship (use noopener noreferrer with target="_blank")
- download : Forces the browser to download the linked file
- title : Tooltip text shown on hover

```
<!-- External link -->
<a href="https://www.google.com">Go to Google</a>

<!-- Open in new tab -->
<a href="https://www.google.com" target="_blank" rel="noopener">Open</a>

<!-- Internal page link -->
<a href="about.html">About Us</a>

<!-- Jump to a section on same page -->
<a href="#section2">Jump to Section 2</a>

<!-- Email link -->
<a href="mailto:info@example.com">Email Us</a>

<!-- Phone link -->
<a href="tel:+911234567890">Call Us</a>
```

13. What is the use of an iframe tag?

The <iframe> (Inline Frame) tag is used to embed another HTML document or an external web page inside the current page. It creates a rectangular window displaying a completely separate document.

Common uses: embedding YouTube videos, Google Maps, external websites, payment widgets.

Important attributes:

- src : URL of the page to embed
- width/height : Dimensions of the frame
- title : Required for accessibility (screen readers)
- sandbox : Restricts what the embedded content can do
- allowfullscreen : Enables fullscreen mode (for videos)
- loading : Set to lazy for performance

```
<!-- Embed a YouTube video -->
<iframe
  src="https://www.youtube.com/embed/VIDEO_ID"
  width="560" height="315"
  title="Tutorial Video"
  allowfullscreen>
</iframe>

<!-- Embed Google Maps -->
<iframe
  src="https://maps.google.com/maps?q=Delhi&output=embed"
  width="600" height="400"
  loading="lazy">
</iframe>
```

14. What is the use of a span tag? Explain with example.

The tag is a generic inline container with no default styling or semantic meaning. Its purpose is to group a piece of inline content so it can be styled with CSS or manipulated with JavaScript, without disturbing the surrounding text flow.

It only becomes useful when combined with class, id, or style attributes.

Difference from <div>: is inline (used inside text/paragraphs), while <div> is block-level (used to wrap sections of content).

```
<!-- Styling part of a sentence -->
<p>The price is <span style="color: red; font-weight: bold;">$99</span> only.</p>

<!-- Using a CSS class -->
<p>Status: <span class="badge-active">Active</span></p>

<!-- JavaScript manipulation -->
<p>Logged in as: <span id="username">Guest</span></p>
<script>
  document.getElementById('username').textContent = 'John Doe';
</script>
```

15. How to insert a picture into a background image of a web page?

To display a background image on the page and place a foreground image on top of it, we use CSS for the background and a regular tag for the overlay, positioned using CSS.

The background image is set as a CSS property (background-image) on the body or a container. It is purely decorative.

The foreground tag is a real HTML element placed on top. It can be positioned anywhere using CSS position properties.

background-size: cover -- scales the image to fill the whole area

position: absolute -- positions the img relative to the parent container

```

<style>
  body {
    background-image: url('background.jpg');
    background-size: cover;
    background-repeat: no-repeat;
    background-position: center;
    min-height: 100vh;
    position: relative;
    margin: 0;
  }
  .overlay-image {
    position: absolute;
    top: 50%;
    left: 50%;
    transform: translate(-50%, -50%);
    width: 200px;
  }
</style>

<body>
  
</body>

```

16. How are active links different from normal links?

HTML anchor (<a>) links have four states, each styled using CSS pseudo-classes:

1. Normal Link (a:link) : A link not yet visited. Default colour is blue with underline.
2. Visited Link (a:visited) : A link the user has already clicked. Default colour is purple.
3. Hover (a:hover) : When the mouse pointer is over the link.
4. Active Link (a:active) : The moment the link is being clicked (mouse held down). Default colour is red.

The order of writing these in CSS matters. The correct order is remembered by the phrase LoVe HAtE: Link, Visited, Hover, Active.

```

a:link { color: blue; text-decoration: underline; }
a:visited { color: purple; }
a:hover { color: orange; text-decoration: none; }
a:active { color: red; font-weight: bold; }

```

17. What are the different tags to separate sections of text?

HTML provides several tags to divide and structure content:

Visual Separators:

- <hr> : Draws a horizontal line to visually divide sections.
-
 : Inserts a line break within text.

- <p> : Creates a paragraph block with automatic vertical spacing.

Semantic / Structural Elements (HTML5):

- <div> : Generic block container (no meaning, just structure).
- <section> : Thematic grouping of related content, usually with a heading.
- <article> : Self-contained, independently distributable content.
- <aside> : Content related but tangential to the main content.
- <header> : Introductory content for a page or section.
- <footer> : Footer of the page or an element.
- <main> : The primary content area of the document.
- <nav> : A section containing navigation links.
- <h1>-<h6> : Heading tags that visually and semantically divide content.

18. What is SVG?

SVG stands for Scalable Vector Graphics. It is an XML-based language for defining two-dimensional vector graphics that can be embedded directly in HTML.

Unlike raster image formats such as JPG or PNG (which are made of pixels), SVG graphics are described mathematically using shapes, paths, and coordinates. This means they scale to any size without becoming blurry or pixelated.

Key features:

- Resolution independent: Sharp and clear at any size or zoom level.
- Styled with CSS: Colours, borders, and effects can be controlled via CSS.
- Scriptable: Each shape is a DOM element, accessible by JavaScript.
- Accessible: Text inside SVG is readable by search engines and screen readers.
- Best suited for: icons, logos, charts, diagrams, and UI graphics.

SVG vs Canvas: Canvas draws pixels (raster) and is better for dynamic rendering like games. SVG is better for scalable, static or interactive graphics.

```
<svg width="200" height="100" xmlns="http://www.w3.org/2000/svg">

  <!-- Rectangle -->
  <rect x="10" y="10" width="80" height="60" fill="steelblue"/>

  <!-- Circle -->
  <circle cx="150" cy="50" r="40" fill="tomato"/>

  <!-- Text -->
  <text x="20" y="90" font-size="14">Hello SVG</text>

</svg>
```

19. What is the difference between HTML and XHTML?

HTML (HyperText Markup Language) is the standard markup language for web pages. XHTML (eXtensible HyperText Markup Language) is a stricter, XML-based version of HTML.

Key Differences:

1. Tag Case:

HTML : Case-insensitive. <P> and <p> are both valid.
XHTML : All tags must be lowercase. Only <p> is valid.

2. Closing Tags:

HTML : Some closing tags are optional.
XHTML : Every tag must be properly closed.

3. Self-closing Tags:

HTML :
 is fine.
XHTML : Must be written as

4. Attribute Quoting:

HTML : Quotes around values sometimes optional.
XHTML : All attribute values must be in quotes.

5. Boolean Attributes:

HTML : <input disabled> is valid.
XHTML : Must be <input disabled="disabled">

6. Error Handling:

HTML : Forgiving parser. Page renders even with errors.
XHTML : Returns error if document is not valid XML.

7. MIME Type:

HTML : text/html
XHTML : application/xhtml+xml

```
<!-- HTML: permissive -->
<INPUT TYPE=TEXT>      <!-- uppercase, unquoted, no closing tag -->

<!-- XHTML: strict -->
<input type="text" />   <!-- lowercase, quoted, self-closed -->
```

20. What are logical and physical tags in HTML?

HTML tags are categorised into two types based on whether they describe appearance or meaning.

Physical Tags (Presentational):

These tags instruct the browser on how to visually render the text. They only affect appearance and carry no meaning about the content's importance.

 - Bold (visual only)
<i> - Italic (visual only)
<u> - Underline

- <s> - Strikethrough
- <small> - Smaller text
- <big> - Larger text (deprecated in HTML5)
- <tt> - Monospace font (deprecated)

Logical Tags (Semantic):

These tags describe the meaning or role of the content. They tell the browser and assistive technologies what the content is, not just how it looks.

- - Important content (renders bold)
- - Emphasized content (renders italic)
- <code> - Computer code
- <abbr> - Abbreviation
- <cite> - Citation or reference
- <dfn> - Definition term
- <var> - Variable in math or code

Best Practice: Always prefer logical/semantic tags over physical tags.

Semantic HTML improves accessibility, helps screen readers, and benefits SEO.

```
<!-- Physical: only changes appearance -->
<b>Warning</b>

<!-- Logical: communicates meaning -->
<strong>Warning</strong>

<!-- Both look bold in the browser, but <strong> tells
      screen readers and search engines it is important -->
```